

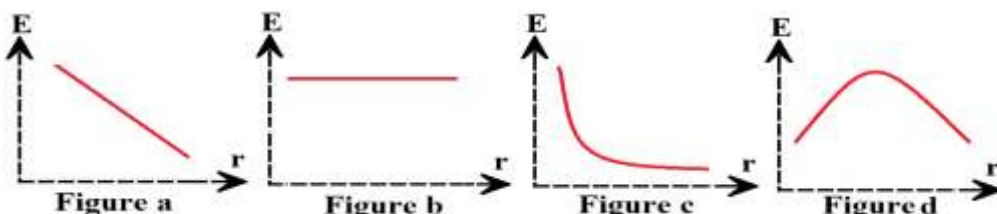
Engineering Physics II	Faculty of Engineering	Date: 31 /8/2025
First Test		
Name:	Reg. No.:	

Section I: Answer all the following by putting true (T) or false (F): **6 Marks**

1) The electric flux is directed in the direction of the electric field lines		
2) A man pressure on the ground decreases as the man sleeps on the ground		
3) Astronaut can not use the straw to drink		

Section II: Answer all the following by selecting the correct choice: **10 Marks**

1) If the electric filed makes an angle $\theta = 30^\circ$ with the normal. The electric flux is:				
a) EA	b) $3EA/2$	c) $EA/2$	d) 0	e) None of these
2) Two spheres, one with radius r and the other with radius 2r, surround a positive charged. The ratio of the number of field lines through the larger sphere to the smaller sphere is				
a) 1/2	b) 0	c) 2/1	d) 4	e) None of these
3) Which of the following figures represent the relation between the electric field at a distance r from a point charge:				



a) Figure b	b) Figure c	c) Figure a	d) Figure d	e) None of these
4) A pressure change With depth				
a) Exponentially	b) Constant	c) Linearity	d) $P_A > P_B$	e) None of these
5) If $\vec{E} = 2\hat{i} + 3\hat{j} - \hat{k}$. Which of the following area is parallel to E:				
a) $2\hat{i} - \hat{j} + \hat{k}$	b) $3\hat{i} - 2\hat{j} + \hat{k}$	c) $-\hat{i} + 3\hat{j} - \hat{k}$	d) $-\hat{i} - 3\hat{j} - \hat{k}$	e) None of these

Section III: Answer all the following by selecting the correct choice

Full and clear steps are required

24 Marks

Constant		
Electron: $m = 9.11 \times 10^{-31}$ kg	$K_e = 9 \times 10^9$ N.m ² /C ²	Gravity: $g = 10$ m/s ²
Proton: $m = 1.67 \times 10^{-27}$ kg	$G = 6.67 \times 10^{-11}$ N.m ² /kg ²	Water: $\rho_f = 1000$ kg/m ³

1) A solid sphere of a radius 40 cm has a total positive charge 26 μ C uniformly distributed through its volume, The electric flux in N.m²/C at 400 mm is:

a) 1020.7	b) - 920.1	c) 920.1	d) -1020.7	e) None of these
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2) Two spheres each of 2 cm radius as shown below, the repulsive force between the spheres is 1200 mN. If the total charge of the spheres is $7 \mu\text{C}$ then the charge on each sphere is:

a) $2 \mu\text{C}$ and $5 \mu\text{C}$	b) $3 \mu\text{C}$ and $4 \mu\text{C}$	c) $1 \mu\text{C}$ and $6 \mu\text{C}$	d) $6 \mu\text{C}$ and $1 \mu\text{C}$	e) None of these
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3) A charge q is placed in the middle of between two charges having 6 C . For system equilibrium, q is:

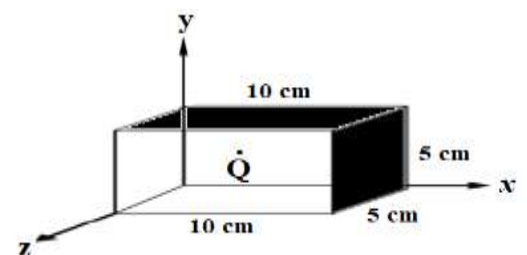
a) 2 C	b) -1.5 C	c) 3 C	d) -2 C	e) None of these
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4) A safety life ring made from plastic with density 396.5 kg/m^3 is fully immersed to keep 30 % of a man volume above the water. The ratio of man to the ring volumes is:

Hint: man density 1070 kg/m^3

a) 1.631	b) 1.661	c) 1.613	d) 1.642	e) None of these
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5) A charge Q is enclosed by a closed shape as shown in figure. If $Q = 26.55 \times 10^{-12} \text{ C}$. Determine the electric flux of the right and top faces of the shape (shaded area)



a) $0.8 \text{ N.m}^2/\text{C}$	b) $1.05 \text{ N.m}^2/\text{C}$	c) $0.9 \text{ N.m}^2/\text{C}$	d) $0.667 \text{ N.m}^2/\text{C}$	e) None of these
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6) During a rainy day the atmospheric pressure reduces to 0.98 atm but it is 1 atm the house having a door with dimension $3 \text{ m} \times 2 \text{ m}$. The force required to open the door is:

a) 198000 N	b) 12000 N	c) 2000 N	d) 200000 N	e) None of these
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